WHAT IS A DERIVATIVE?

A derivative is an instrument whose value depends on, or is derived from, the value of another asset.

Examples: futures, forwards, swaps, options

WHY DERIVATIVES ARE IMPORTANT

Derivatives play a key role in transferring risks in the economy

The underlying assets include stocks, currencies, interest rates, commodities, debt instruments, electricity prices, insurance payouts, the weather, etc

Many financial transactions have embedded derivatives

HOW DERIVATIVES ARE TRADED

On exchanges such as the Chicago Mercantile Exchange (CME)

In the over-the-counter (OTC) market where traders working for banks, fund managers and corporate treasurers contact each other directly

THE OTC MARKET

Largely unregulated prior to 2008

Banks acted as market makers quoting bids and asks

Master agreements usually defined how transactions between two parties would be handled

Some transactions were cleared through central counterparties (CCPs). A CCP stands between the two sides to a transaction in the same way that an exchange does

SINCE 2008

OTC market has become regulated. Objectives:

- Reduce systemic risk
- Increase transparency

In the U.S and some other countries, standardized OTC products must be traded on swap execution facilities (SEFs) which are electronic platforms similar to exchanges

CCPs must be used to clear standardized transactions between financial institutions in most countries

All trades must be reported to a central repository

HOW DERIVATIVES ARE USED

To hedge risks

To speculate (take a view on the future direction of the market)

To lock in an arbitrage profit

To change the nature of a liability

To change the nature of an investment without incurring the costs of selling one portfolio and buying another

FORWARD PRICE

The forward price for a contract is the delivery price that would be applicable to the contract if were negotiated today (i.e., it is the delivery price that would make the contract worth exactly zero)

The forward price may be different for contracts of different maturities

TERMINOLOGY

The party that has agreed to buy has what is termed a long position

The party that has agreed to sell has what is termed a short position

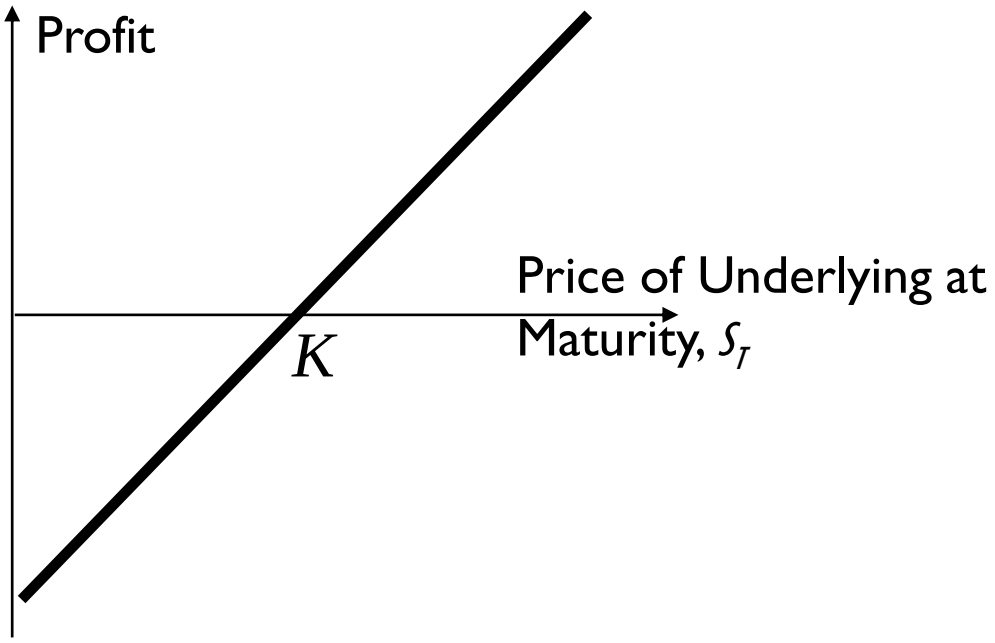
EXAMPLE

On May 21, 2020, the treasurer of a corporation enters into a long forward contract to buy £1 million in six months at an exchange rate of 1.2230

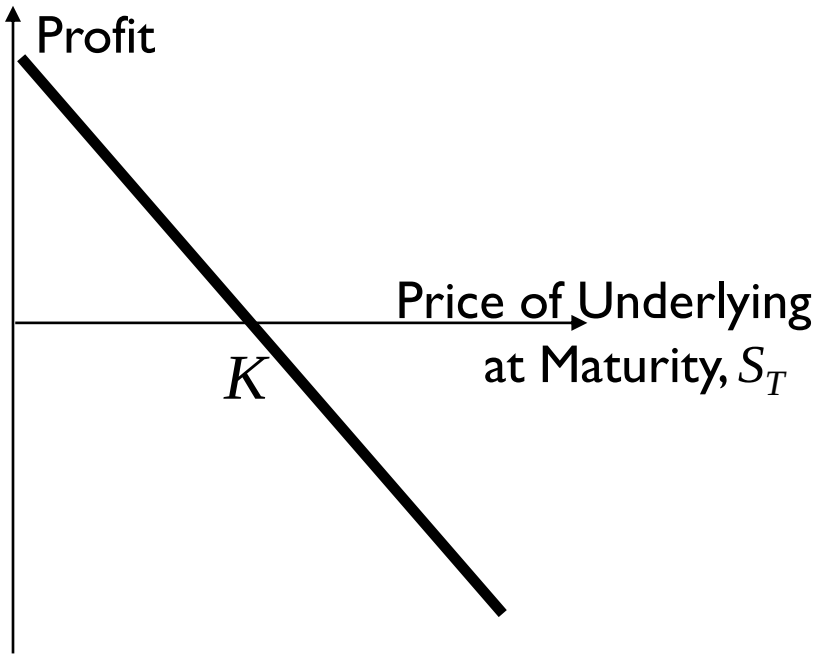
This obligates the corporation to pay $1,223,000 for £1 million on November 21, 2020

What are the possible outcomes?

LONG FORWARD POSITION



SHORT FORWARD POSITION



FUTURES CONTRACTS

Agreement to buy or sell an asset for a certain price at a certain time

Similar to forward contract

Whereas a forward contract is traded OTC, a futures contract is traded on an exchange

EXCHANGES TRADING FUTURES

CME Group (formed when Chicago Mercantile Exchange and Chicago Board of Trade merged)

InterContinental Exchange

B3 (Brazil)

Tokyo Financial Exchange (Tokyo)

and many more (see list at end of book)

EXAMPLES OF FUTURES CONTRACTS

Agreement to:

- Buy 100 oz. of gold @ US$1800/oz. in December
- Sell £62,500 @ 1.2500 US$/£ in March
- Sell 1,000 bbl. of oil @ US$40/bbl. in April

FORWARD PRICE OF A NON-DIVIDEND PAYING STOCK

If the spot price is S and the forward price for a contract deliverable in T years is F, then

$$F = S\,(1+r\,)^T$$

where r is the 1-year (domestic currency) risk-free rate of interest.

In our examples, S = 60, T = 1, and r =0.05 so that

$$F = 60(1+0.05) = 63$$

OPTIONS

A call option is an option to buy a certain asset by a certain date for a certain price (the strike price)

A put option is an option to sell a certain asset by a certain date for a certain price (the strike price)

AMERICAN VS EUROPEAN OPTIONS

An American option can be exercised at any time during its life

A European option can be exercised only at maturity

OPTIONS VS FUTURES/FORWARDS

A futures/forward contract gives the holder the obligation to buy or sell at a certain price

An option gives the holder the right to buy or sell at a certain price

TYPES OF TRADERS

Hedgers

Speculators

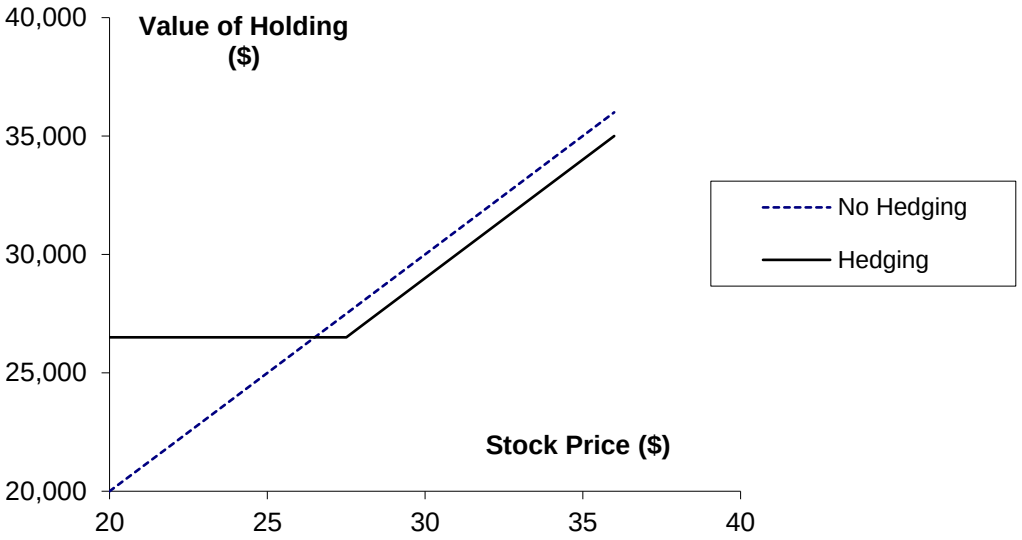
Arbitrageurs

HEDGING EXAMPLES

A US company will pay £10 million for imports from Britain in 3 months and decides to hedge using a long position in a forward contract

An investor owns 1,000 shares currently worth $28 per share. A two-month put with a strike price of $27.50 costs $1. The investor decides to hedge by buying 10 contracts

VALUE OF SHARES WITH AND WITHOUT HEDGING



SPECULATION EXAMPLE

An investor with $2,000 to invest feels that a stock price will increase over the next 2 months. The current stock price is $20 and the price of a 2-month call option with a strike of 22.50 is $1

What are the alternative strategies?

ARBITRAGE EXAMPLE

A stock price is quoted as £100 in London and $120 in New York

The current exchange rate is 1.2300

What is the arbitrage opportunity?

DANGERS

Traders can switch from being hedgers to speculators

It is important to set up controls to ensure that trades are using derivatives in for their intended purpose

HEDGE FUNDS

Hedge funds are not subject to the same rules as mutual funds and cannot offer their securities publicly.

Mutual funds must

- disclose investment policies,
- make shares redeemable at any time,
- limit use of leverage

Hedge funds are not subject to these constraints.

Hedge funds use complex trading strategies and are big users of derivatives for hedging, speculation and arbitrage

EXAMPLES OF HEDGE FUND STRATEGIES

Long/Short Equities

Convertible Arbitrage

Distressed Securities

Emerging Markets

Global Macro

Merger Arbitrage